02323 Introduction to Statistics

Lecture 1: Introduction: Descriptive Statistics and Python

DTU Compute
Technical University of Denmark
2800 Lyngby – Denmark

IntroStat team


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- 02402 Statistics (Polytechnical Foundation)
- 02323 Introduction to Statistics
- 02403 Introduction to Mathematical Statistics

Fall 2026: Statistics Courses

- 02323 lectures in English: Friday 8-10, by Md Saifuddin Khalid
- 02402 lectures in Danish: Tuesdays 8-10, by Pernille Yde Nielsen

Students taking the course 02402 can attend lectures in 02323 (and vice versa) if they prefer the language of the lecture

NOTE: There are small differences between the two courses (2-way ANOVA and matrix formulation of linear regression are not covered in 02323).

Fall 2026: Statistics Courses

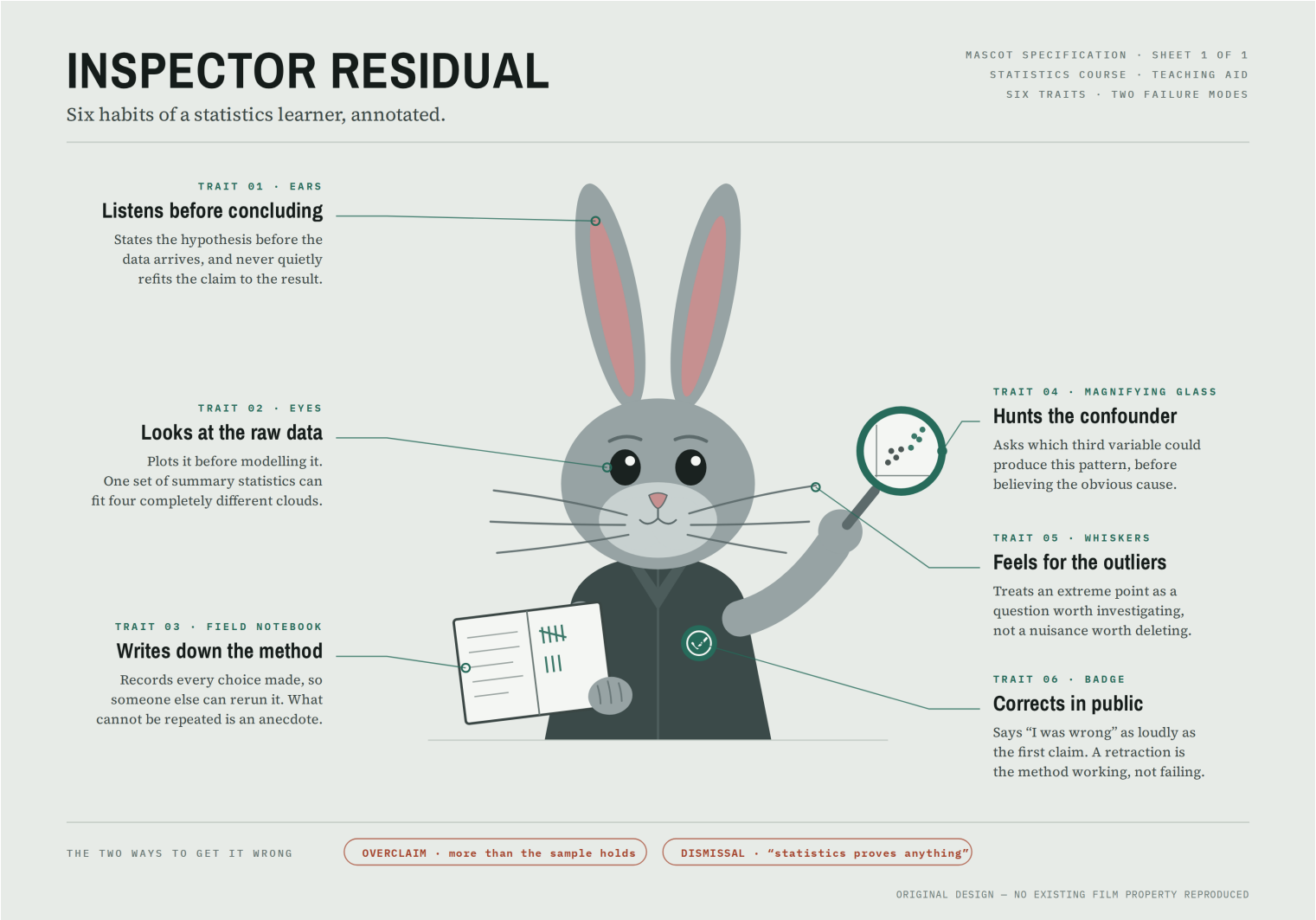


Figure: Khalid's Course Mascot

Agenda

1. Practical Info: Locations and Roles
2. Practical info: Assessment and Feedback
3. Introduction and Motivation
4. Descriptive Statistics
   - Mean and median
   - Percentiles and quantiles
   - Variance and standard deviation
   - Sample covariance and Sample correlation
5. Graphical Methods: Data Visualization
6. Software: Python in Visual Studio Code

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Teaching

- Lectures: Fridays 08:00-10:00 Location: Building 306 - Auditorium 33 (Room Capacity: 194)
- Exercises: Fridays 10-12
  - Building 324 - Room 40 (Capacity 60) - Teaching assistant: Alberte Bech Elholm
  - Building 324 - Room 50 (Capacity 36) (First Three Weeks) - Teaching assistant: Anna Siricová
  - Building 324 - Room 60 (Capacity 60) - Teaching assistant: Ask Gram Franck
  - Building 324 - Ground Floor - Study Areas 003 (Capacity 18), 004 (Capacity 24), and 005 (Capacity 18)
  - Python Support: Building 302

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Exam and Projects

- Exam
  - Saturday, 19 December 2026
  - 4-hour multiple choice
  - Starting from Fall 2026, **the exam is only offered in English**.
  - Written works of reference are permitted. Strictly, the calculator models TI30XS/TI30XB are allowed.
- Mandatory projects
  - 2 optional projects should be submitted via Learn for competence development.
  - For each project, choose one of four topics.
  - Via Learn, hand in project 1 in week 7 and project 2 in week 10 of the course.
  - If you've already submitted projects, then you may revise and submit to get feedback.

Practical info: Book yourself for routinized learning activities

- Generic weekly agenda
  - Before teaching: 2 hours, Read relevant sections in the book
  - Lectures: 2 hours, curriculum of the week
  - Exercises: 2 hours, exercises and online quizzes
  - After teaching: 3 hours, exercises, previous exam questions & quizzes on the website
- Teaching material
  - Available under *Agendas* on course website
  - Lecture slides and Python code might be updated shortly before each lecture (remember to refresh browser)



Practical Info: Content and Queries

- Website: 02323.compute.dtu.dk
  - Agendas: Teaching plan, syllabus, exercises and solutions (English), slides (Danish and English), scripts & quiz, previous exams and solutions
  - Book
  - Projects (&Deadlines) - Submit via Learn
  - Podcasts of previous years' lectures (OBS: no recordings using Python)
  - Quizzes
- DTU Learn
  - Announcements
  - Projects - description and submission
- EdStem Discussion (Join Now!)
  - Q&A `https://edstem.org/eu/join/3xAC9c`

Python

In addition to paper/pencil & calculator, we must use Python to calculate statistical tasks.





As a trial, you will get access to www.DataCamp.com, where you can refresh or expand your Python skills.

The use of DataCamp is voluntary.



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Preamble

Statistics is a mathematical science of collecting, describing, analyzing, and interpreting data.
You will extract knowledge and ***learn from observed data.***

Probability is a branch of mathematics that deals with the description and analysis of chance.
You will derive knowledge and ***learn from a theoretical model***

The fields are difficult to separate and methods from both fields are used commonly together in engineering.

Et fælles mål: A common goal: Describe and understand random (stochastic) variation and uncertainties quantitatively!

History & Application of Statistics in Various Fields

There are many exciting research areas within both applied and theoretical statistics.

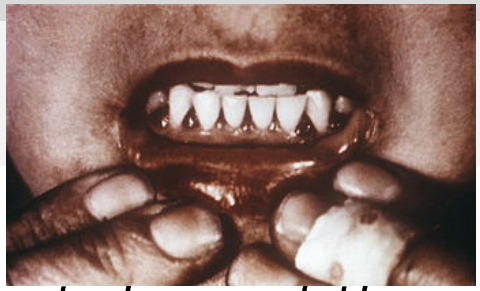
Statistics are used in many areas, for example:

- Clinical studies and epidemiology
- Production, planning and quality control
- Analysis of laboratory data and experiments
- Data Science and Artificial Intelligence (AI)

Al-Khwarizmi and Al-Kindi in the 9th Century

- **Al-Khwarizmi's** (c. 780–850) most famous work, "Kitab **al-Jabr** wal-Muqabala" (The Compendious Book on Calculation by Completion and Balancing), is where the term "**algebra**" originates. He is the "father of algebra and **algorithm**" - The foundation for modern mathematics and statistical methods. Invented the systematic use of Hindu-Arabic numerals.
- **Al-Kindi** (c. 801–873) considered a pioneer in statistics for developing frequency analysis, the earliest known form of statistical inference. He developed the method of frequency analysis in his manuscript Risalah fi Istikhraj al-Mu'amma (Manuscript on Deciphering Cryptographic Messages), which used the statistical frequency of letters to break codes.

Clinical Trial by James Lind in 1747



In 1747, James Lind, a Scottish naval surgeon, conducted one of the earliest recorded clinical trials to tackle scurvy, a disease rampant among sailors. Aboard the HMS Salisbury, Lind selected 12 sailors afflicted with scurvy and divided them into six groups of two. Each group received different dietary supplements, including cider, vinegar, seawater, and oranges with lemons. Lind meticulously observed and compared the outcomes, finding that the sailors who consumed citrus fruits showed the most significant recovery. This experiment marked the first application of statistical principles in medicine, as Lind systematically compared different treatments to identify the most effective one. This led the British Navy to adopt lemon juice in sailors' diets, effectively eradicating scurvy. Lind's approach laid the foundation for modern clinical trials and evidence-based medicine.

(See `http://en.wikipedia.org/wiki/James_Lind`).

Statistics in Modern Epidemiology by John Snow

John Snow, in 1854 during a cholera outbreak in London, employed a statistical approach by mapping cholera cases in the Soho district and identifying a pattern. Snow's analysis revealed that the majority of cases clustered around a public water pump on Broad Street.
He hypothesized that contaminated water, not miasma (the prevailing theory), was the source of the outbreak. Snow's method of plotting cases on a map and calculating the distribution of disease concerning water sources provided clear evidence supporting his theory. This use of statistical mapping and spatial analysis in public health marked a significant shift in understanding disease transmission, laying the foundation for modern epidemiology. His work emphasized the importance of data in identifying and controlling outbreaks, a principle that remains central to public health today.



Why should an engineer or scientist learn statistics?

- Handling Uncertainty and Variability: To quantify the measurement variability to ensure designs remain robust under real-world conditions.
- Data-Driven Decision Making: Statistical tools (e.g., confidence intervals, hypothesis testing) to draw valid conclusions from experiments and simulations.
- Quality Control and Improvement: To monitor manufacturing processes in real-time, reducing defects and improving efficiency.
- Predictive Maintenance: To predict machine failures before they occur, reducing downtime and maintenance costs.
- Algorithm Development: Statistics are foundational for training machine learning models, signal processing, and optimizing network performance.

Common Fallacies and Biases

Statistics can be counterintuitive, and our brains need to be trained in statistical thinking to avoid making a series of common fallacies. *Even well-educated, professional statisticians make simple mistakes*.

Some typical biases (systematic distortions) in statistics are:

- Survivorship bias: **Ignoring failures**, focusing on successes.
- Selection bias: Distorted results due to **sample non-representativeness**.
- OVB (Omitted-variable bias): **Missing variables** skew results and conclusions.

Course's Learning Objectives Covered Today

- Estimate and interpret simple summary statistics, such as mean, standard deviation, variance, median, and quartiles as well as correlation
- Apply simple graphical techniques, including histograms, **qq-normal plots (*not today*)**, and box plots
- Use Python and interpret its output
- Recognize core statistical terms

The course content in broad terms

A large part of the course covers:

1. Formulation of models
2. Calculation of confidence intervals
3. Conducting hypothesis tests
4. determining whether various conditions are *statistically significant*



in different contexts and setups.

Probability theory will be our primary tool.
The course should also prepare you for advanced courses in experimental design, time series analysis, quality control, probability theory, statistical modelling, data analysis, machine learning, and artificial intelligence.
Limitations: (1) Small datasets, (2) Simplified scenarios lack real-world complexities and context.

Basics of Statistics

Statistics can generally be divided into two parts:

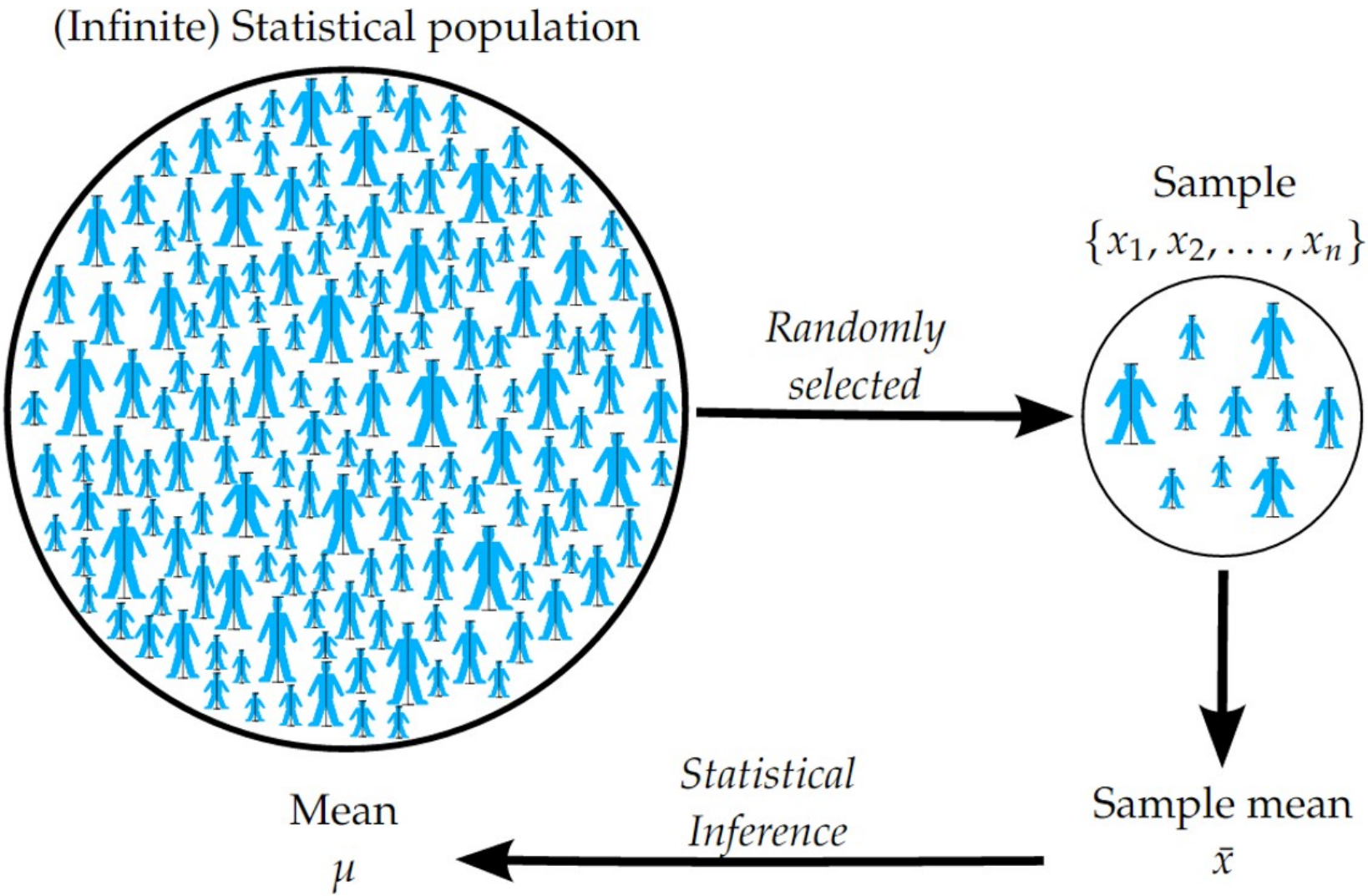
- Descriptive statistics
- Inferential statistics

Statistics typically involves analyzing a *sample* taken from a *population*.

From the sample, we generally make statements about the population.

It is therefore important that the sample is *representative* of the population. *For most of the course, we will simply assume that the samples are representative.*

Statistics


(Infinite) Statistical population
Randomly selected
Sample
$\{x_1, x_2, \ldots, x_n\}$
Sample mean
$\bar{x}$
Statistical Inference
Mean
μ

The Population

The word "Population" is used in statistics in a slightly different way than it is used in everyday language.

A statistical population does not have to consist of people — it can also be a population of objects or events.

The statistical population can be concretely defined (e.g. "all students at DTU throughout 2025") or it can be more abstract and possibly infinitely large (e.g. "all conceivable cases of allergies").

It is not always clearly defined what the population is, but it is always a good idea to try to define the population.

A representative sample

The sample will always be clearly defined, since it is this that our concrete **data** concerns.

When we perform *statistical inference*, we make general statements about the population based on observations in the sample. We therefore *assume* that the tendencies observed in the sample can be generalised to the population. It is therefore important that the sample is *representative* of the population.

A sample is representative when it is drawn completely at random — i.e., all units in the population have the *same* probability of being included in the sample.

For most of the course, we will simply assume that the samples are representative.

An example with sample and population

Let us say, we want to know something about well-being among students at DTU. We therefore create a questionnaire and distribute it to the 54 students who show up to the course "Discrete Mathematics" on Thursday morning (this course is in the 1st semester of the Software Technology program). The questionnaire contains a number of questions about both academic and social well-being. Not all students want to participate, but we receive responses from 42 students.

Questions:
What is the statistical population?
Is the sample representative of the population we want to investigate?
How could we have made a better sample?

(Write your own notes here)

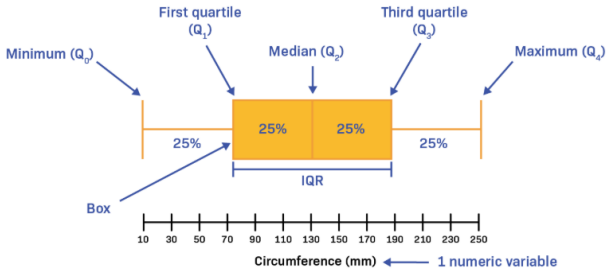
Overview

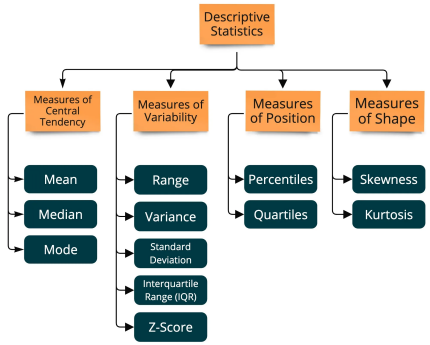
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Descriptive Statistics

To describe and summarize **data**, using: (1) Tabular method, (2) Graphical method, (3) Numerical method

Class Interval	Tally Marks	Frequency	Cumulative Frequency (cf)
20 - 30	||	2	2
30 - 40	||||	4	6
40 - 50	||||		6	12
50 - 60	||	2	14
60 - 70	||||		6	20
70 - 80	||||	4	24





Different types of data

- Numerical data
  - Quantitative values measured on a scale (cm, kr, ..)
  - Discrete values (0/1, counts)
- Qualitative/Categorical data
  - Groupings (male/female, before/after, red/yellow/blue)
  - Possibly ranked groups (low/medium/high, "to a low degree"/"to a high degree")

Qualitative data are often "translated" into binary values using 0/1 (so-called "dummy variables").

The General Setup

There is an underlying population from which a representative sample with n observations has been drawn.

The sample is usually represented by a vector

$$x = (x_1, x_2, \ldots, x_n).$$

The sorted sample is then

$$(x_{(1)}, x_{(2)}, \ldots, x_{(n)}),$$

where $x_{(1)}$ denotes the smallest observation and $x_{(n)}$ denotes the largest observation.

Example: Sample of 20 observations

A sample with 20 observations has the following values:

$$41, 44, 37, 38, 36, 39, 38, 47, 32, 43, 36, 39, 49, 35, 40, 35, 45, 36, 44, 38$$

Here are the sample values in sorted order:

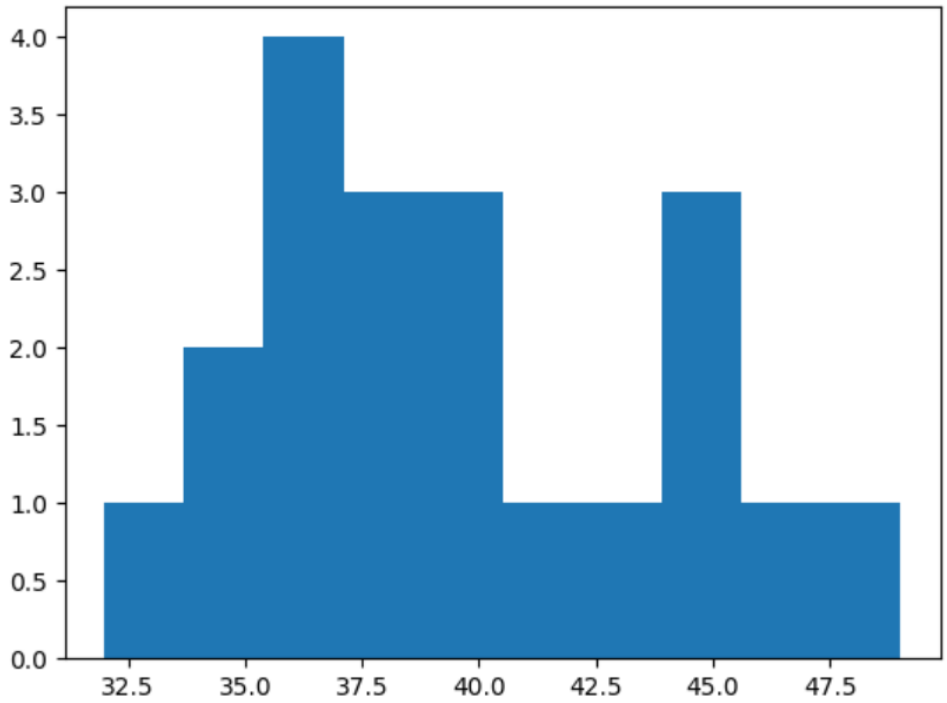
$$32, 35, 35, 36, 36, 36, 37, 38, 38, 38, 39, 39, 40, 41, 43, 44, 44, 45, 47, 49$$

How can the sample be described?

(Write your own notes here)

Example: visualization of data

We can also visualize the sample. For example with a histogram:



$$32, 35, 35, 36, 36, 36, 37, 38, 38, 38, 39, 39, 40, 41, 43, 44, 44, 45, 47, 49$$

What does the histogram actually show?

Kahoot!

(x1)

Summary statistics

We use *summary statistics* to summarize and describe data

- Measures of position (e.g. centrality)
  - e.g.: mean ($\bar{x}$), median and quantile
- Measures of dispersion
  - e.g.: range, inter-quartile range (IQR), variance (s^2) and standard deviation (s)
- Measures of relation
  - e.g.: covariance and correlation

Note the difference between, e.g., the *(sample)* mean $\bar{x}$ and the *(population)* mean μ.

Mean, Definition 1.4

The mean value is a key number which indicates the centre of gravity or centering of the data.

The sample mean (*average*):

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

We say that $\bar{x}$ is an *estimate* of the population mean.

Mean or *Average*)

Assume we have a sample with a total of n observations:

$$\{x_1, x_2, x_3, x_4, x_5, x_6, x_7, \ldots, x_n\}$$

Definition 1.4 Sample mean

The sample mean is the sum of observations divided by the number of observations

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \qquad (1\text{-}1)$$

Sometimes this is refereed to as the *average*.

32 35 35 36 36 36 37 38 38 38 39 39 40 41 43 44 44 45 47 49

$$\bar{x} = 39.6 \quad (n = 20)$$

Median, Definition 1.5

The median is also a key number indicating the center of the data.

In some cases, for example in the case of extreme values, the median is preferable to the mean.

Sample median:
The middle observation (of the sorted data) or the average of the two middle observations (of the sorted data), depending on whether the sample has an even or odd number of observations.



Example: Student heights

- **Sample:** Student heights in cm, $n = 5$.

$$(x_1, x_2, x_3, x_4, x_5) = (185, 184, 194, 180, 182)$$

- **Mean:**

$$\bar{x} = \frac{1}{5}(185 + 184 + 194 + 180 + 182) = 185$$

- **Median:**
  - First order the data: $180, 182, 184, 185, 194$.
  - Then choose the third/middle number (n uneven): 184
- If a person with height 235 cm is added to the data:
  - *Mean:* 193
  - *Median:* 184.5

Median

Definition 1.5 Median

Order the n observations $x_1, \ldots, x_n$ from the smallest to largest: $x_{(1)}, \ldots, x_{(n)}$. The median is defined as:

- If n is odd the median is the observation in position $\frac{n+1}{2}$:

$$Q_2 = x_{(\frac{n+1}{2})}. \tag{1-2}$$

- If n is even the median is the average of the two observations in positions $\frac{n}{2}$ and $\frac{n+2}{2}$:

$$Q_2 = \frac{x_{(\frac{n}{2})} + x_{(\frac{n+2}{2})}}{2}. \tag{1-3}$$

The reason why it is denoted with Q_2 is explained below in Definition 1.8.

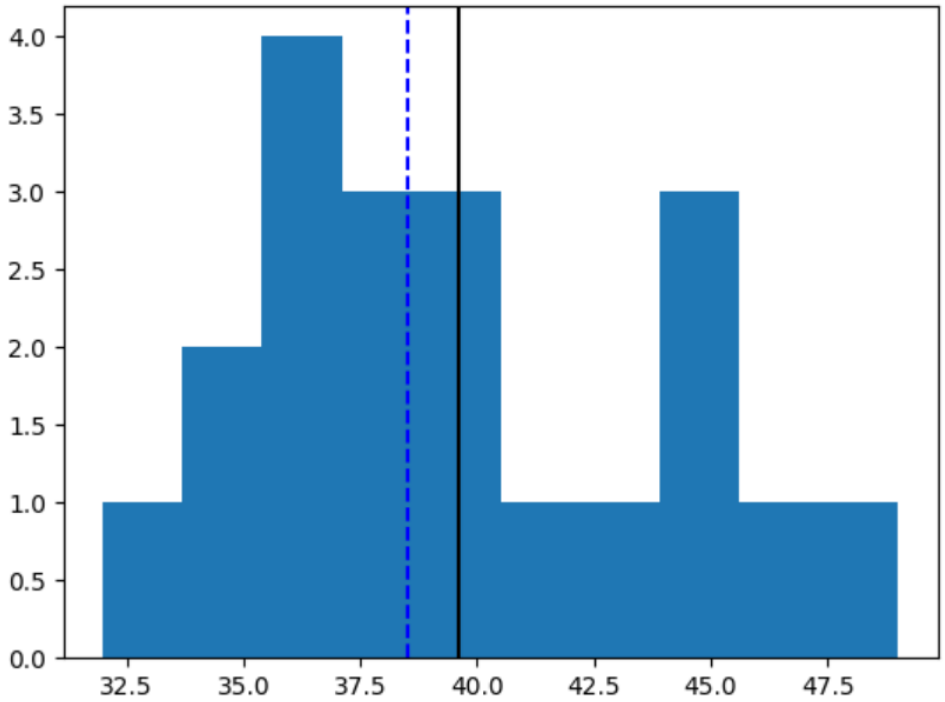
32 35 35 36 36 36 37 38 38 38 | 39 39 40 41 43 44 44 45 47 49

$$median = 38.5$$

Mean and Median

Mean and median are not the same (but they will often lie close to each other).

Here is the example from earlier along with the mean (black) and median (blue):



Percentiles and quantiles

The median is the value that divides the data into two halves. More generally, we may compute *percentiles*, e.g.:

- $0, 25, 50, 75, 100$ % percentiles and/or
- $0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100$ % percentiles

Note:

- The median is the 50% percentile.
- The 25, 50, 75 % percentiles are often referred to as the first, second and third quartiles, and denoted $Q1$, $Q2$, and $Q3$, respectively.
- Inter Quartile Range (IQR): $Q3 - Q1$

Quantiles or *Percentiles*)

Definition 1.7 Quantiles and percentiles

The p *quantile* also called the $100p\%$ quantile or $100p$'th *percentile*, can be defined by the following procedure: [a]

1. Order the n observations from smallest to largest: $x_{(1)}, \ldots, x_{(n)}$
2. Compute pn
3. If pn is an integer: average the pn'th and $(pn+1)$'th ordered observations. Then the p quantile is

$$q_p = \left(x_{(np)} + x_{(np+1)}\right)/2 \tag{1-4}$$

4. If pn is a non-integer: take the "next one" in the ordered list. Then the p'th quantile is

$$q_p = x_{(\lceil np \rceil)}, \tag{1-5}$$

where $\lceil np \rceil$ is the *ceiling* of np, that is, the smallest integer larger than np

Example: Student heights

- **Sample:** *Ordered* student heights in cm.

$$(x_{(1)}, x_{(2)}, x_{(3)}, x_{(4)}, x_{(5)}) = (180, 182, 184, 185, 194)$$

- **Lower quartile, Q1:**
  - Establish that $np = 1.25$, as $p = 0.25$ and $n = 5$.
  - The smallest integer larger than np is 2.
  - $Q1 = x_{(2)} = 182$.
- **Upper quartile, Q3:**
  - Establish that $np = 3.75$, as $p = 0.75$ and $n = 5$.
  - The smallest integer larger than np is 4.
  - $Q3 = q_{0.75} = x_{(\lceil 3.75 \rceil)} = x_{(4)} = 185$.
- **IQR:**
  - $Q3 - Q1 = 185 - 182 = 3$.

Percentiles: Example

The 50% percentile (the median) illustrated in the example from earlier:

32 35 35 36 36 36 37 38 38 38 | 39 39 40 41 43 44 44 45 47 49

That is, the median here is 38.5.

The 10% percentile illustrated:

32 35 | 35 36 36 36 37 38 38 38 39 39 40 41 43 44 44 45 47 49

That is, the 10% percentile here is 35.

If a percentage falls "between" the number of observations in the data, the nearest value is used. See formula on the previous slide.

For example, the 33% percentile is 37.

Range and IQR

Definition 1.15 Range

The *range* of the sample is

$$\text{Range} = \text{Maximum} - \text{Minimum} = Q_4 - Q_0 = x_{(n)} - x_{(1)}. \tag{1-9}$$

The Inter Quartile Range (IQR) is the middle 50% range of data defined as

$$IQR = q_{0.75} - q_{0.25} = Q_3 - Q_1. \tag{1-10}$$

NOTE: Minimum and Maximum are also called Q0 and Q4 here

Example, Exam 02323 December 2025

Exercise IV

An industrial engineer is analyzing the production efficiency scores (on a 0–100 scale) of a machine across 30 consecutive days (the measurements can be assumed to be independent). The goal is to assess how consistently the machine performs and to detect possible outliers of unusually high or low efficiency. The quartiles for the daily efficiency scores are as follows:

$$Q_1 = 72, \quad Q_2 = 78, \quad Q_3 = 88.$$

Question IV.1 (6)

Which of the following statements about the Inter Quartile Range (IQR) is correct?

1 ☐ The IQR measures the total spread between the smallest and largest efficiency scores in the dataset.

2 ☐ The IQR is calculated as $Q_1 - Q_3 = 72 - 88 = -16$, which represents the left-skewness of the data.

3 ☐ The IQR is $Q_3 - Q_1 = 88 - 72 = 16$, representing the range of the middle 50% of efficiency scores.

4 ☐ The IQR equals the median ($Q_2 = 78$), which divides the dataset into two equal halves.

5 ☐ The IQR is used only for normally distributed data to calculate the standard deviation.

6 ☐ Don't know / No answer

Example, Exam 02323 December 2025

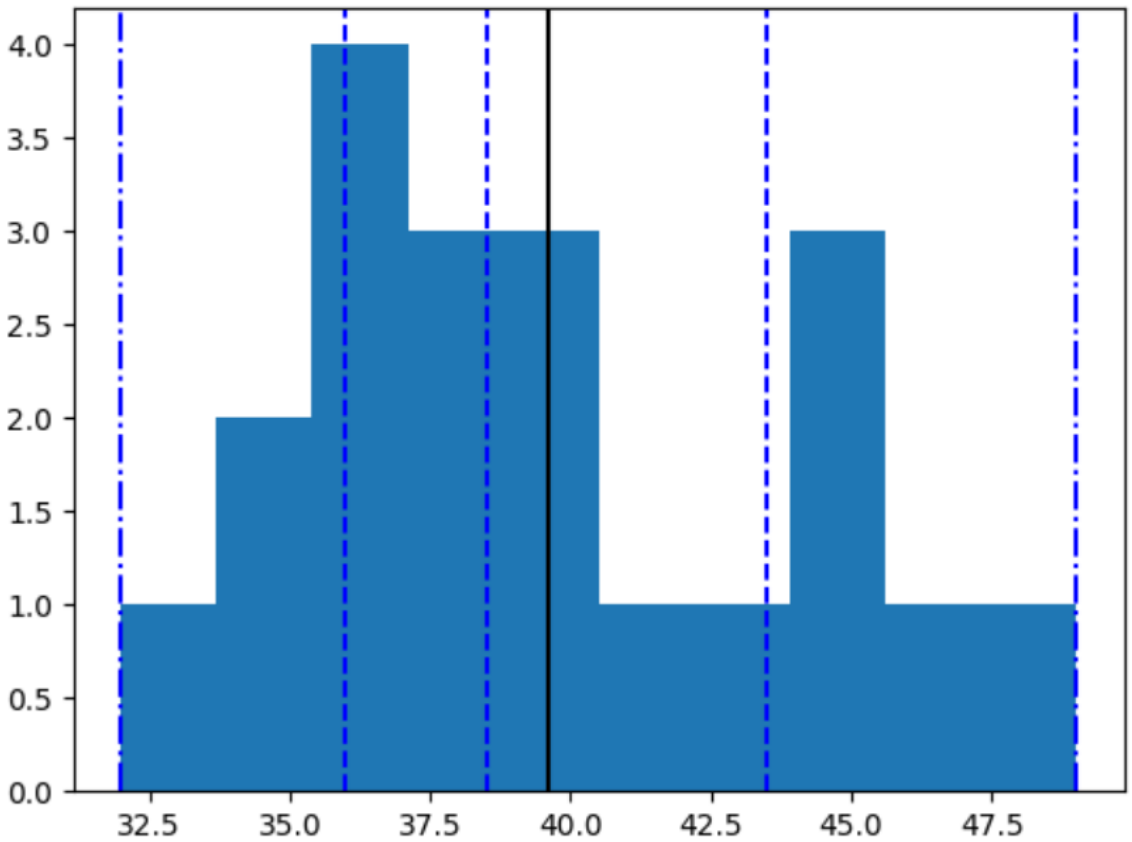
The Interquartile Range (IQR) quantifies the variability of the central 50% of the data and is defined as: $IQR = Q_3 - Q_1$. Here,

$IQR = 88 - 72 = 16.$

This means that on typical days, the machine's efficiency lies between 72 and 88 points. A large IQR indicates inconsistent production performance, while a smaller IQR implies the machine operates with stable efficiency. Outlier days can be identified using:

Lower bound: $Q_1 - 1.5 \times IQR = 72 - 24 = 48$,
Upper bound: $Q_3 + 1.5 \times IQR = 88 + 24 = 112$.
Since efficiency scores above 100 are impossible, any day below 48 would be considered an outlier of poor performance.

Visualization of min, Q1, Q2, mean, Q3 and max



Kahoot!

Sample variance and Sample standard deviation, Definition 1.10

The variance and the standard deviation indicate the dispersion ("spread") of the data:

- Variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

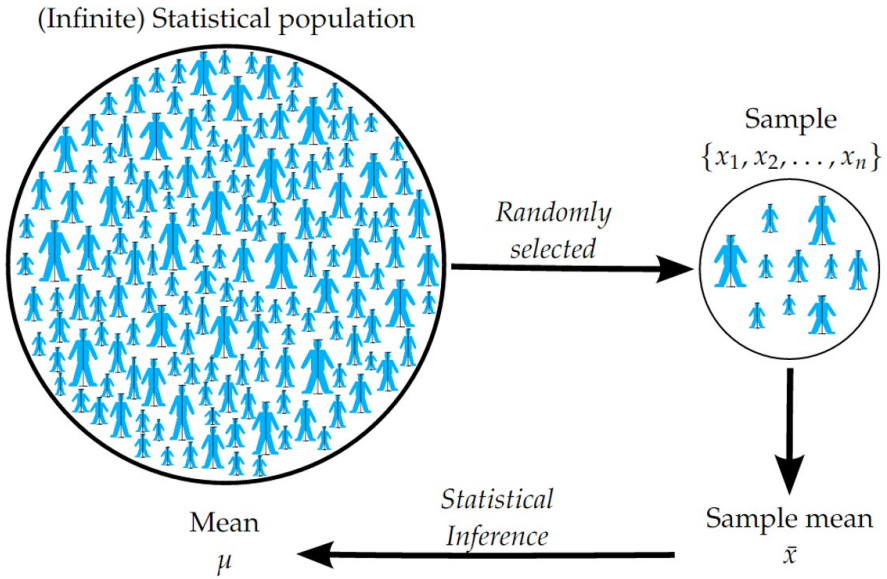
- Standard deviation

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Why $n-1$?

Why $n-1$?

When we define *"Sample variance"* and *"Sample standard deviation"*, it is to obtain the best *estimate* of the population variance and standard deviation.



Therefore, we divide by $(n-1)$ instead of n.
(called, Bessel's correction, named after Friedrich Bessel)

Be careful, there are different definitions out there! However, this is the most common definition of *"Sample variance"* and *"Sample standard deviation"*.

Example on Dispersion/Spread: Student heights

- **Sample:** Student heights in cm, $n = 5$.

$$(x_1, x_2, x_3, x_4, x_5) = (185, 184, 194, 180, 182)$$

- **Variance:**

$$s^2 = \frac{1}{4}((185-185)^2 + (184-185)^2 + \cdots + (182-185)^2) = 29$$

- **Standard deviation:**

$$s = \sqrt{29} = 5.385$$



The coefficient of variation, Definition 1.12

The standard deviation and the variance are key numbers for absolute variation.

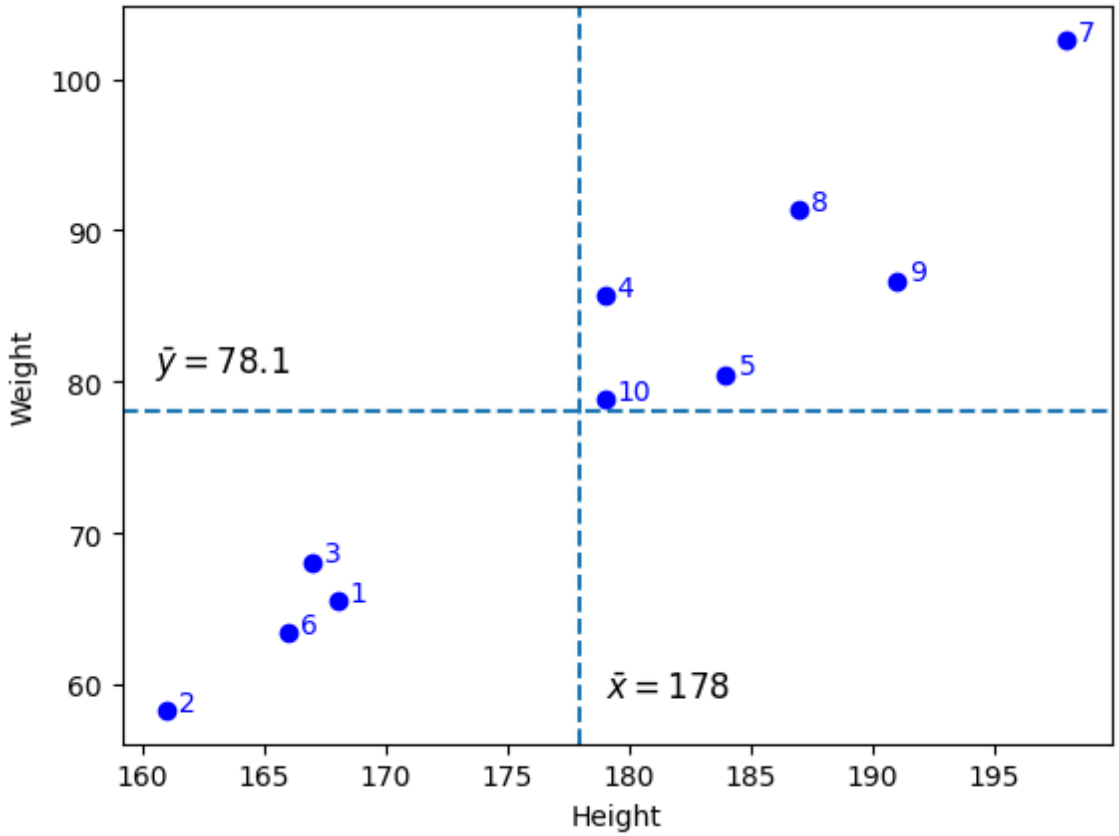
If it is of interest to compare variation between different data sets, it might be a good idea to use a *relative* key number.

Coefficient of variation:

$$V = \frac{s}{\bar{x}} \tag{1}$$

Sample covariance and sample correlation

Observation nr.:	1	2	3	4	5	6	7	8	9	10
Højde (cm) - (x_i)	168	161	167	179	184	166	198	187	191	179
Vægt (kg) - (y_i)	65.5	58.3	68.1	85.7	80.5	63.4	102.6	91.4	86.7	78.9



Sample Covariance and correlation, Definitions 1.18 & 1.19

The sample covariance is given by

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

The sample correlation coefficient is given by

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{s_{xy}}{s_x \cdot s_y}$$

where s_x and s_y are the sample standard deviations for x and y respectively.

Sample covariance and correlation

Student	1	2	3	4	5	6	7	8	9	10
Heights (x_i)	168	161	167	179	184	166	198	187	191	179
Weights (y_i)	65.5	58.3	68.1	85.7	80.5	63.4	102.6	91.4	86.7	78.9
$(x_i - \bar{x})$	-10	-17	-11	1	6	-12	20	9	13	1
$(y_i - \bar{y})$	-12.6	-19.8	-10	7.6	2.4	-14.7	24.5	13.3	8.6	0.8
$(x_i - \bar{x})(y_i - \bar{y})$	126.1	336.8	110.1	7.6	14.3	176.5	489.8	119.6	111.7	0.8

$$
\begin{aligned}
s_{xy} &= \frac{1}{9}(126.1 + 336.8 + 110.1 + 7.6 + 14.3 + 176.5 + 489.8 \\
&\quad + 119.6 + 111.7 + 0.8) \\
&= \frac{1}{9} \cdot 1493.3 \\
&= 165.9
\end{aligned}
$$

Covariance (s_{xy}) gives the direction of a linear relationship, not its strength

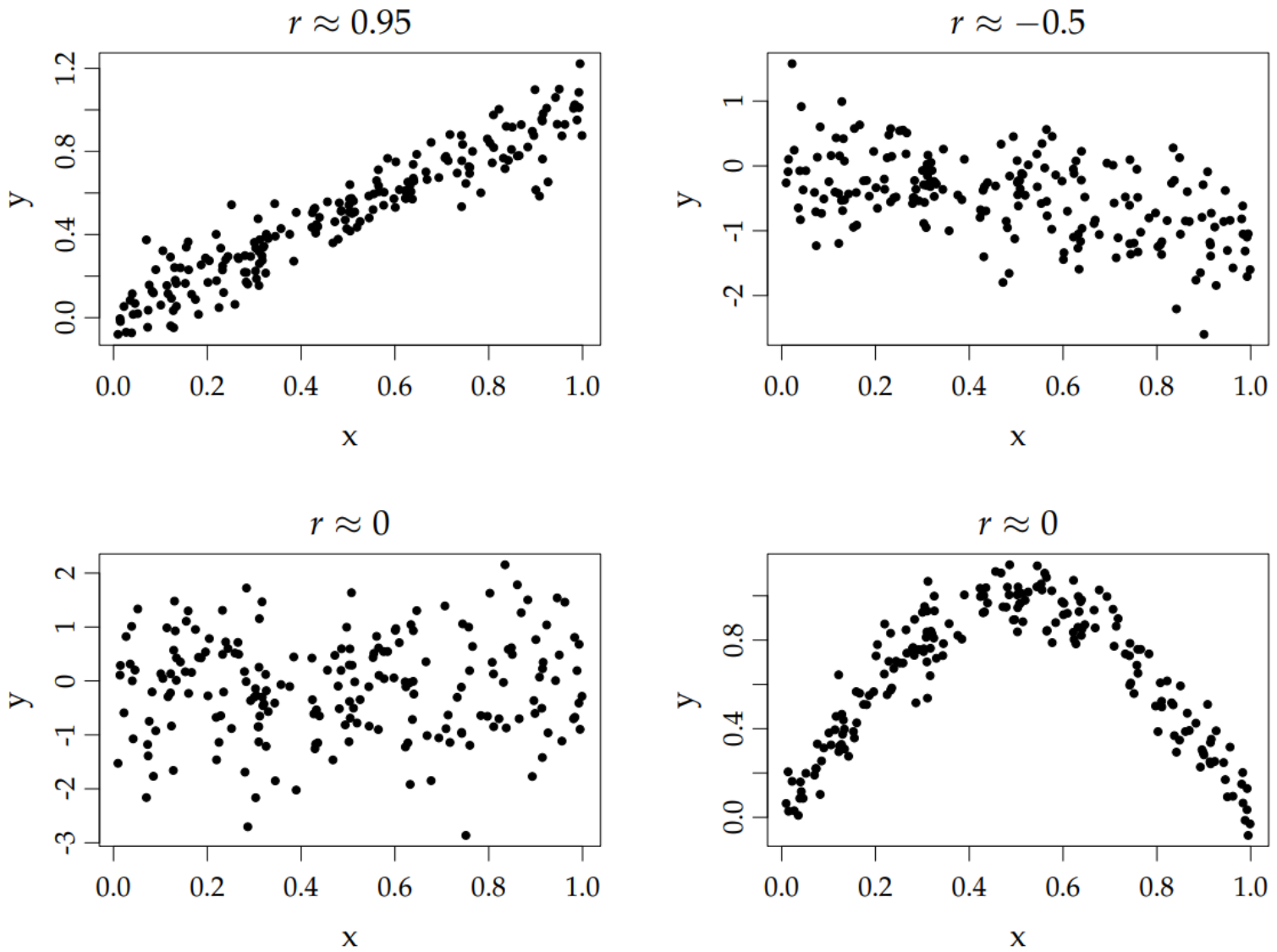
$$s_x = 12.21, \quad \text{and} \quad s_y = 14.07$$

$$r = \frac{165.9}{12.21 \cdot 14.07} = 0.97$$

Properties of Correlation Coefficient

- r is always between -1 and 1: $-1 \leq r \leq 1$.
- r measures the degree of linear relation between x and y.
- $r = \pm 1$ if and only if all points in the scatterplot are exactly on a line.
- $r > 0$ if and only if the general trend in the scatterplot is positive.
- $r < 0$ if and only if the general trend in the scatterplot is negative.

Sample correlation

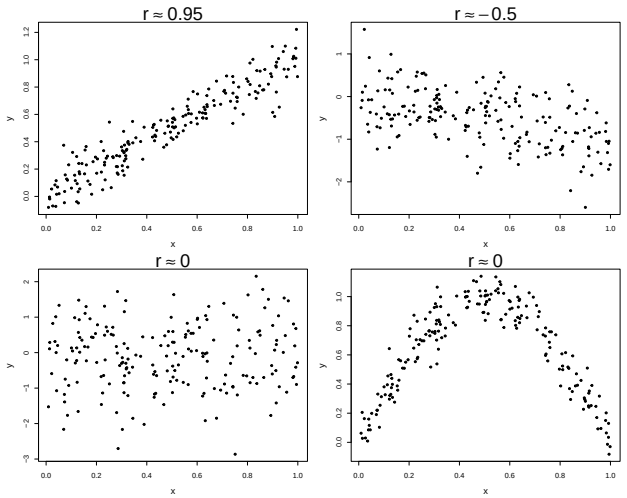


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6. Software: Python in Visual Studio Code

Figures/Tables

- Quantitative data
  - Scatter plot (xy plot)
  - Histogram
  - Cumulative distribution
  - Box plot
- Qualitative/Count data
  - Bar chart
  - Pie chart



Visualization of data is important!
We can make different figures and tables in Python

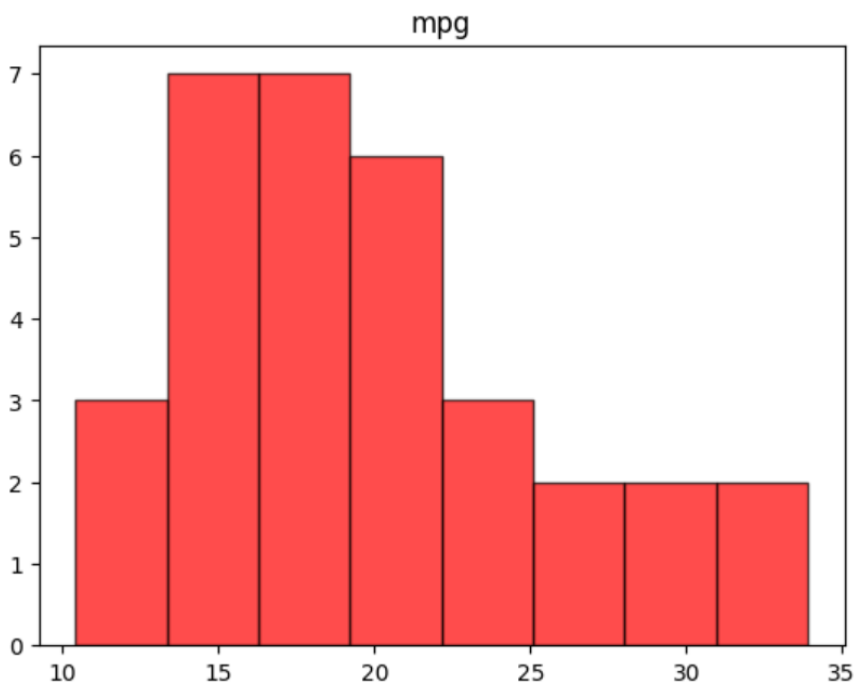
More complex data

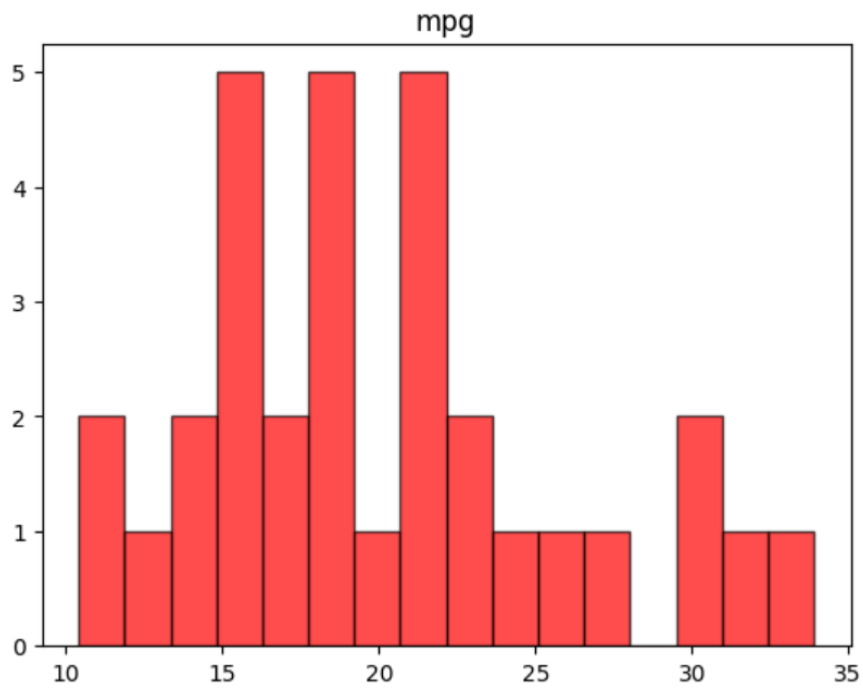
Here is an example with data for several different cars.

model	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21	6	160	110	3.9	2.62
Mazda RX4 Wag	21	6	160	110	3.9	2.875
Datsun 710	22.8	4	108	93	3.85	2.32
Hornet 4 Drive	21.4	6	258	110	3.08	3.215
Hornet Sportabout	18.7	8	360	175	3.15	3.44
Valiant	18.1	6	225	105	2.76	3.46

Histogram

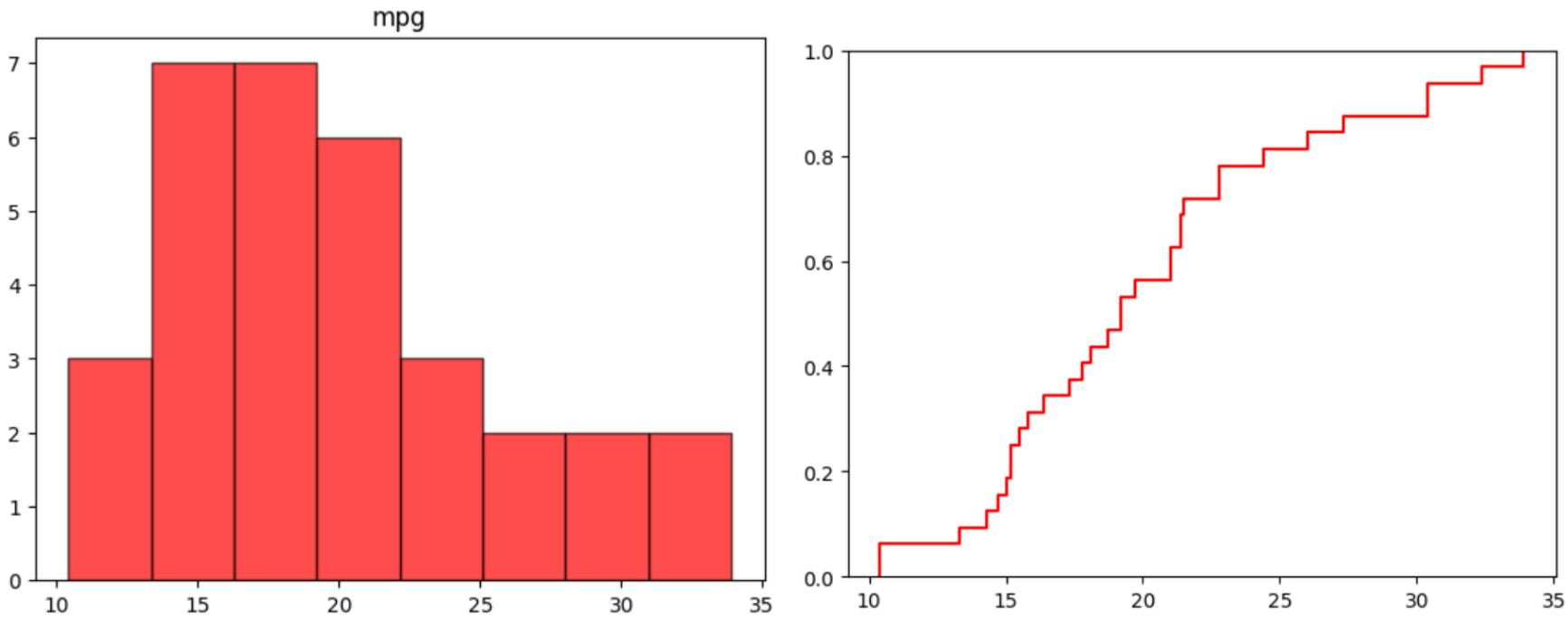
Two histograms for the same data (the column mpg in the car dataset). The difference between the two histograms is solely the choice of "*bin size*".





Cumulative frequency distribution

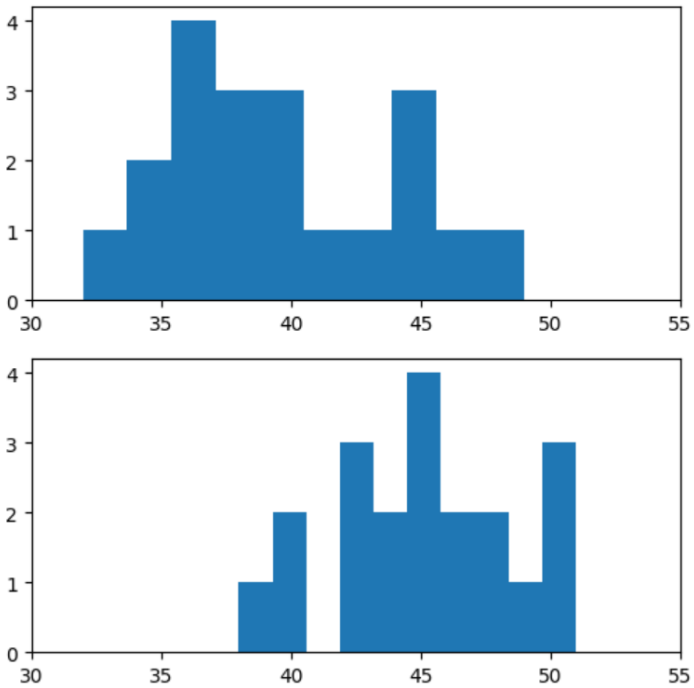
Instead of a histogram, one can visualize the sample with a cumulative plot (also called *empirical cumulative frequency distribution*), which has the advantage that it does not depend on "*bin size*"

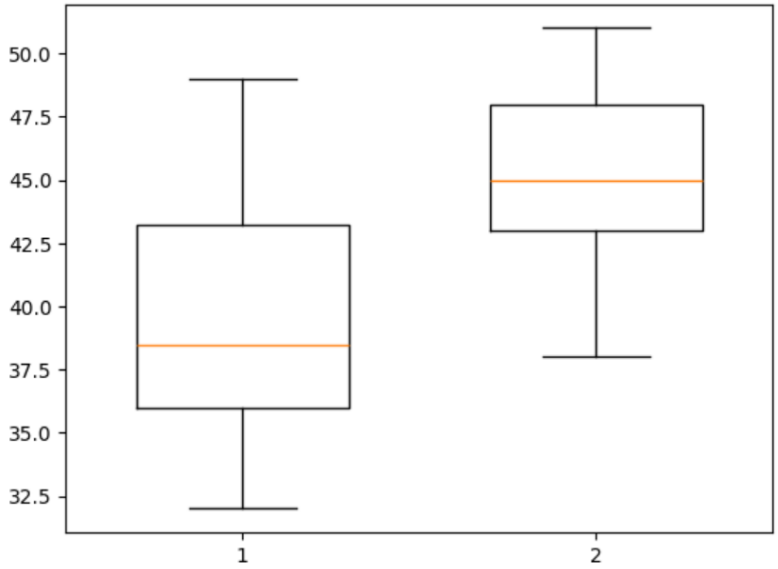


Boxplot

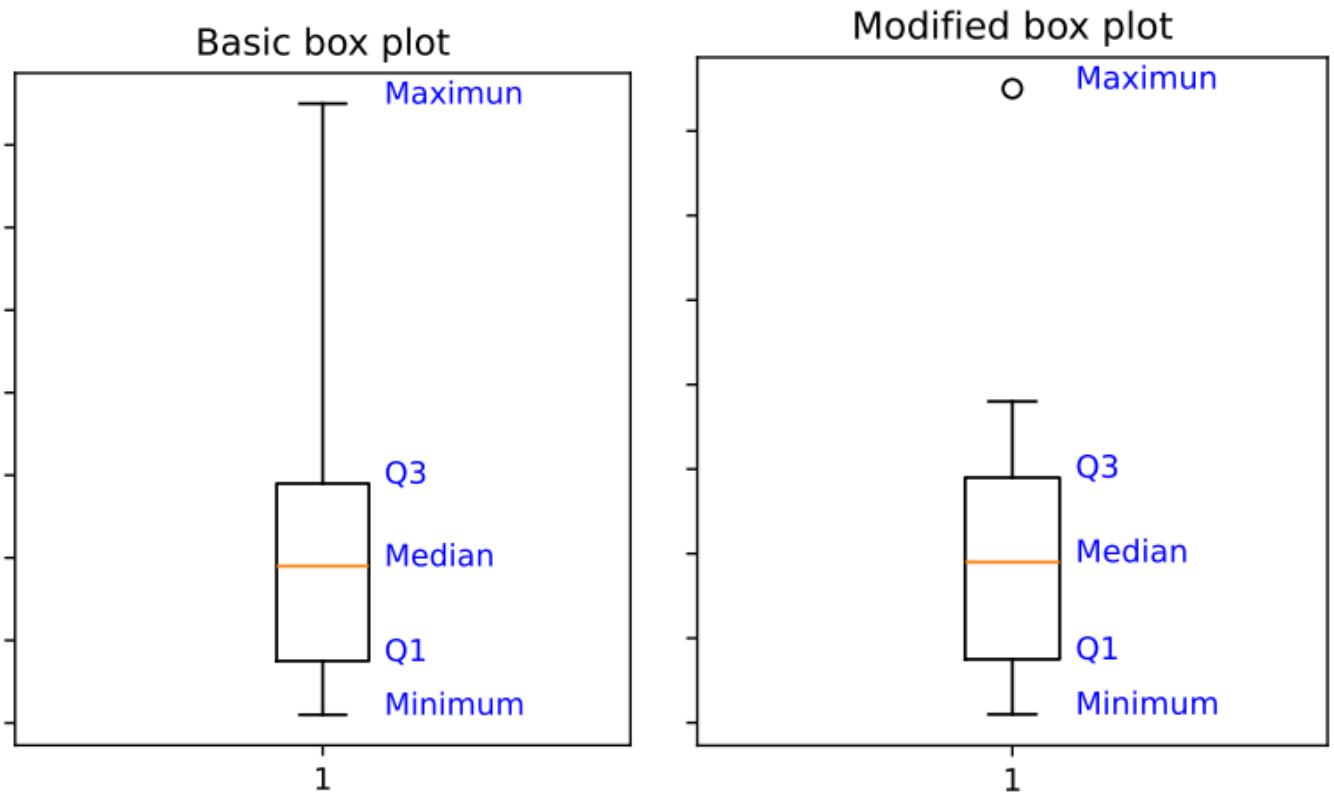
For comparison of multiple samples, boxplots are often used.

The boxplots are (often) drawn with the values along the y-axis instead of along the x-axis (as in the histograms).





How is a boxplot drawn?

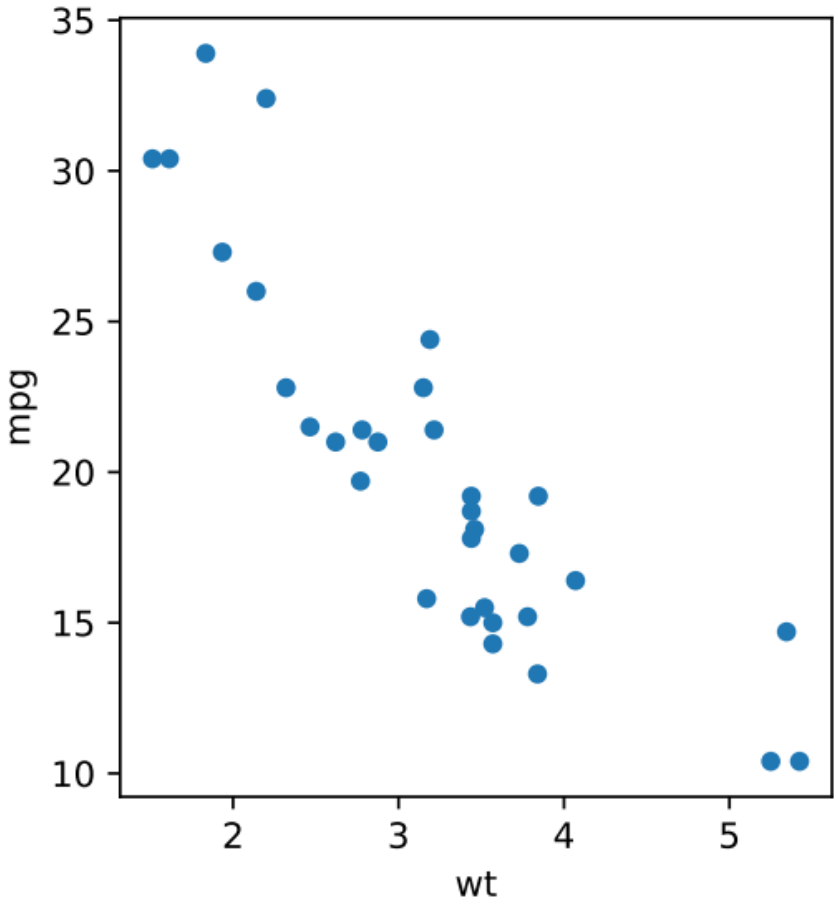


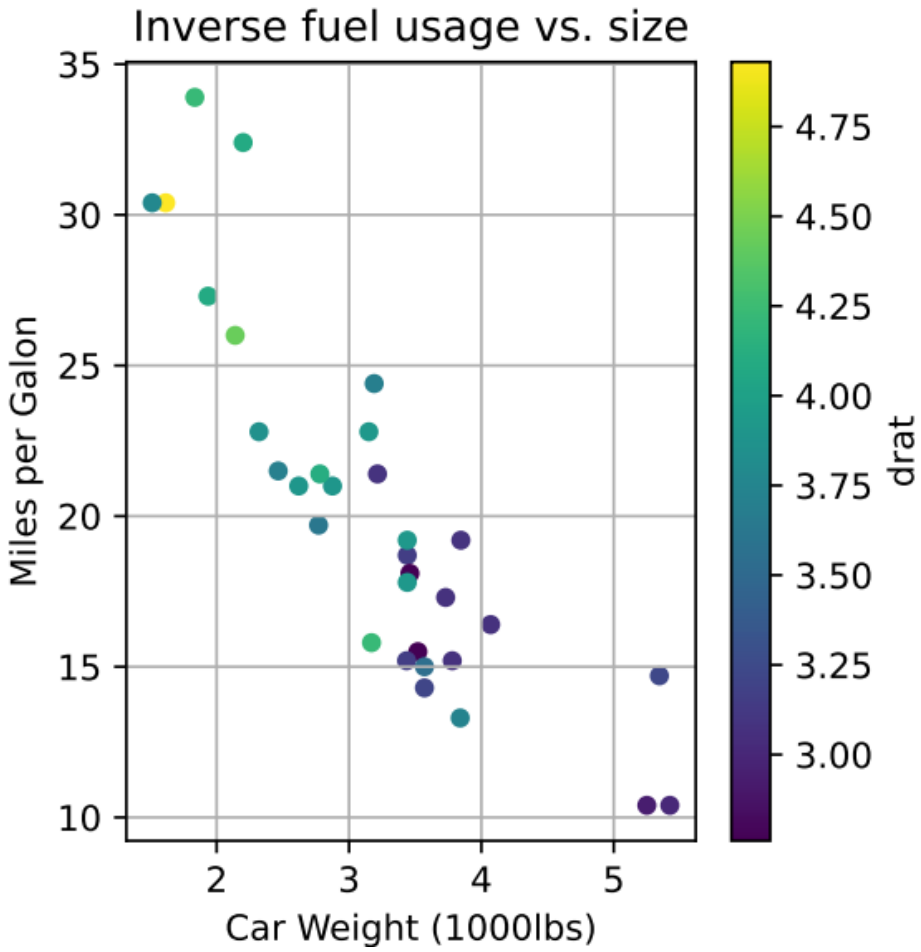
"Whiskers" can be drawn in different ways. The book has 2 definitions.

"Basic boxplot": Whiskers are drawn all the way to the maximum and minimum.

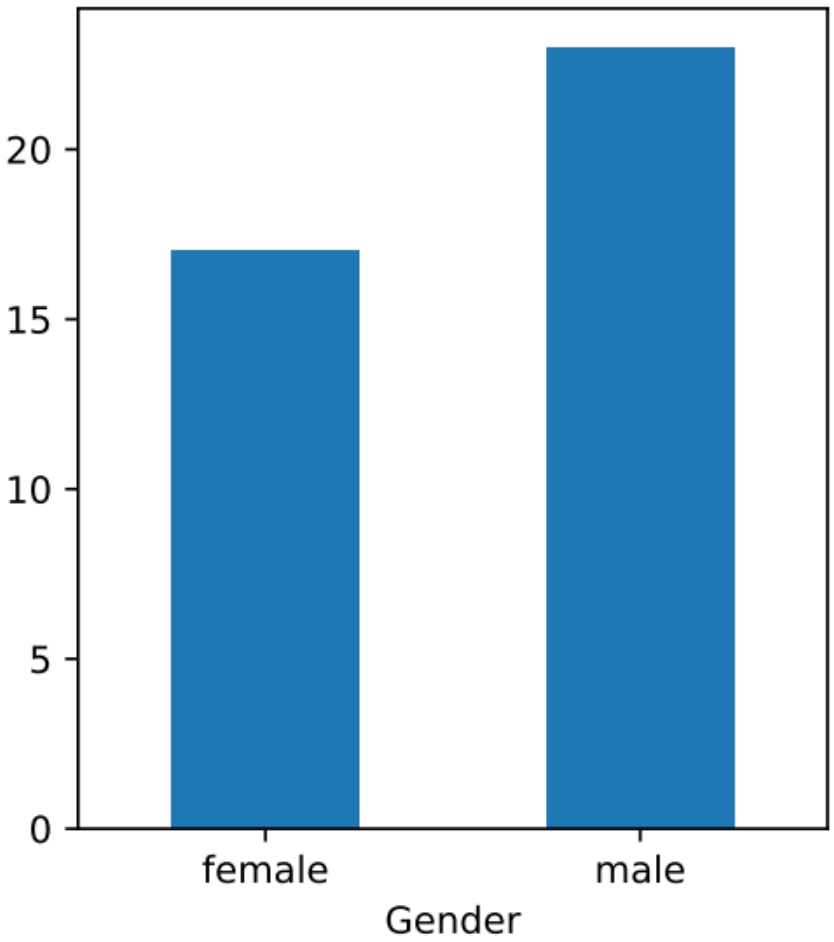
"Modified boxplot": Whiskers are drawn at most out to $1.5 \cdot IQR$ (from Q1/Q3). Values outside this are drawn individually and are called "outliers" or "extreme" values.

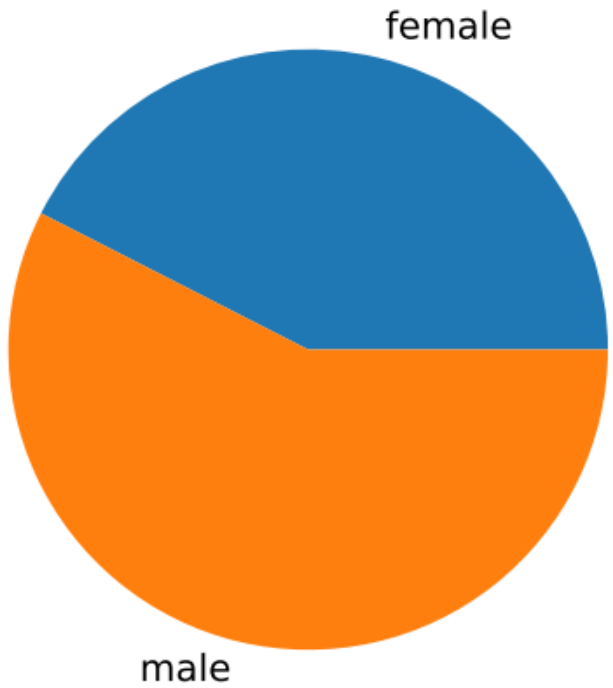
Scatterplot (xy plot)





Bar and pie charts





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Software: Python in Visual Studio Code

- Python: An interpreted, interactive, object-oriented programming language. It provides high-level data structures such as list and associative arrays (dictionaries), dynamic typing and dynamic binding, modules, classes, exceptions, automatic memory management, etc.
- Python: Free, open source, works on all platforms.
- Many Python "libraries" for all kinds of data analysis.
- Introduced in the book & ingeral part of the course.
- Learning by doing.
- Visual Studio Code as code editor using Microsoft Python extension



Software: Python in Visual Studio Code

- In 02323 & 02402 Statistics courses, we use VS Code to edit and run our Python programs.
- We primarily use Jupyter notebooks (.ipynb).
- We expect that you have installed both Python and VS Code and that you can create a Jupyter notebook.
- Help with installation: https://pythonsupport.dtu.dk/

Python abbreviations in 02402/02323

In this course, we use a number of Python "libraries" and **apply the following abbreviations**:

- import numpy as np
- import matplotlib.pyplot as plt
- import pandas as pd
- import scipy.stats as stats
- import statsmodels.api as sm
- import statsmodels.formula.api as smf
- import statsmodels.stats.power as smp
- import statsmodels.stats.proportion as smprop

Python

- Go to today's Python notebook in VS Code



Little more Kahoot



(x8)

Summary from today:

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Exercises

- Exercises start at 1015
- You can find the exercises on the course website's "Agenda" (the exercises, like the book, are only available in English)
  - Start by opening VS Code and creating a new notebook (check that it works by calculating 2+2)
  - Then proceed with the day's exercises (both with and without Python)
  - It's also a good opportunity to discuss the curriculum and ask questions to the teaching assistants
- Most of the teaching assistants speak Danish and all are fluent in English.
- Go to website 02323.compute.dtu.dk for locations of the exercises and names of TAs
- Those having trouble with Python and VS Code can go to room B302.

Exam Question – Percentile and boxplot limit

02323 exam, June 2026

Data: daily minutes on online news, $n = 20$ users (ordered)

$$12, 18, 22, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 100, 110, 125, 140$$

(a) What is the 90th percentile P_{90}?

1. 125
2. 117.5
3. 140
4. 110
5. 100
6. Don't know / No answer

(b) One further observation, 165, is added. Find the upper boxplot limit and decide whether 165 is extreme.

1. Upper limit $= 157.5$; 165 is extreme
2. Upper limit $= 162.5$; 165 is extreme
3. Upper limit $= 162.5$; 165 is not extreme
4. Upper limit $= 172.5$; 165 is not extreme
5. Upper limit $= 157.5$; 165 is not extreme
6. Don't know / No answer

Exam Question – Percentile and boxplot limit

02323 exam, June 2026

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1. Upper limit $= 157.5$; 165 is extreme
2. Upper limit $= 162.5$; 165 is extreme
3. Upper limit $= 162.5$; 165 is not extreme
4. Upper limit $= 172.5$; 165 is not extreme
5. Upper limit $= 157.5$; 165 is not extreme
6. Don't know / No answer

Answers: (a) option 2 | (b) option 4

(a) $np = 0.90 \cdot 20 = 18$ is an integer, so $P_{90} = \frac{1}{2}(x_{(18)} + x_{(19)}) = \frac{1}{2}(110 + 125) = 117.5$.
(b) With $n = 21$: $0.25 \cdot 21 = 5.25 \Rightarrow Q_1 = x_{(6)} = 35$; $0.75 \cdot 21 = 15.75 \Rightarrow Q_3 = x_{(16)} = 90$. Then $\text{IQR} = 55$ and the upper limit is $Q_3 + 1.5\,\text{IQR} = 90 + 82.5 = 172.5$. Since $165 < 172.5$, it is not an extreme observation.

Exam Question – Summary statistics and covariance

02323 exam, May 2025

(a) For the sample $(1, 2, 3, 9, 10)$, which statement is *false*?

1. The lower quartile Q_1 is 2
2. The median is 3
3. The mean is 5
4. The range is 9
5. **The variance is** 16

(b) The covariance between hours studied and exam score is 25.2. Which statement best describes the relationship?

1. There is a weak, positive relationship
2. The relationship is negative
3. **Positive covariance suggests that as hours studied increase, exam scores tend to increase as well**
4. A covariance of 25.2 means the correlation is strong
5. There is no relationship, since covariance alone cannot determine correlation

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Answers: (a) option 5 | (b) option 3

(a) $\bar{x} = \frac{25}{5} = 5$, range $= 10 - 1 = 9$, $q_{0.25} = x_{(\lceil 1.25 \rceil)} = x_{(2)} = 2$, $q_{0.5} = x_{(\lceil 2.5 \rceil)} = x_{(3)} = 3$ – all true. But $s^2 = \frac{16+9+4+16+25}{5-1} = \frac{70}{4} = 17.5$, not 16.
(b) Covariance gives the *direction* of a linear relationship, not its strength: a positive value says the two tend to move together. Judging strength requires the correlation $r = s_{xy}/(s_x s_y)$.

Next week:

- Probability I: Book Chapter Two.